- *Data is formatted correctly.* (Email fields have a valid email format.)
- *Numbers are in a particular range.* (You can't buy -1 copies of a product.)
- *Required values are provided while others are optional.* (Name may be required, but phone number is optional.)
- *Values conform to your business requirements*. (You can't convert a currency to itself, it needs to be converted to a different currency.)

It might seem like some of these can be dealt with easily enough in the browser. For example, if a user is selecting a currency to convert to, don't let them pick the same currency; and we've all seen the "please enter a valid email address" messages.

Unfortunately, although this *client-side validation* is useful for users, as it gives them instant feedback, you can never rely on it, as it will always be possible to bypass these browser protections. It's always necessary to validate the data as it arrives at your web application using *server-side validation*.

WARNING Always validate user input on the server side of your application.

If that feels a little redundant, like you'll be duplicating logic and code between your client and server applications, I'm afraid you're right. It's one of the unfortunate aspects of web development; the duplication is a necessary evil. Fortunately, ASP.NET Core provides several features to try to reduce this burden.

TIP Blazor, the new C# single-page application (SPA) framework, promises to solve some of these problems. For details, see http://mng.bz/9D51 and *Blazor in Action*, by Chris Sainty (Manning, 2021).